

MSCM 2.0 Final Report

Why Translation-Invariant Prime-Shift Operators Don't Encode Arithmetic in k-Tuple Ratio Observables

Status

This document formalizes the complete set of developments, definitions, falsification tests, empirical outcomes, and the mathematical explanation supporting the final conclusion:

In translation-invariant linear models with smooth positive forcing, Hardy-Littlewood k-tuple constants are determined by local sieve admissibility [1,3,4].

The global “prime-shift” operator acts as an **arithmetic-agnostic smoother**, and its effect cancels in the ratio observables studied.

This is a **negative result** for a broad class of operator-based prime models, and it provides a reusable **null-test framework** for future operator proposals.

1. Research question

We test whether a translation-invariant linear operator built from prime-indexed shifts can encode arithmetic structure relevant to the Hardy-Littlewood prime k-tuple correlations [1].

Two hypotheses:

- **H₀ (agnostic smoothing):** Prime-shift dynamics is effectively indistinguishable from random sparse shift dynamics after stability normalization; observed constants are sieve-driven.
 - **H₁ (arithmetic content):** Prime-shift dynamics produces residue-specific distortions and/or spectral signatures not replicated by random-support operators.
-

2. Model definition (MSCM 2.0)

2.1 Ambient space

Work on the finite cyclic group

$$G = \mathbb{Z}/N\mathbb{Z}, \text{ with } f \in \ell^2(G).$$

The periodic setting enables FFT-based diagonalization and reproducible numerics.

2.2 Layer 1: translation-invariant dynamics

Define a circulant convolution operator

$$(Af)(n) = \sum_{k \in G} a(k)f(n-k),$$

with kernel supported on a sparse set of shifts.

Prime-support kernel (nominal model):

$$a(p) = \lambda p^{-\alpha} \text{ for primes } p \leq P, \quad a(k) = 0 \text{ otherwise.}$$

Random-support kernel (null comparator): sample a sparse support on $\{2, \dots, P\}$ with “prime-like” inclusion probabilities proportional to $1/\log n$ (cf. Cramér-style random models [2,5]), then assign the same weight rule λn^α .

Stability normalization: choose λ so that

$$\max_{\xi \in G^*} |\widehat{a}(\xi)| = \rho \in (0,1),$$

where $\widehat{a}$ is the discrete Fourier transform (DFT) symbol.

2.3 Forcing and fixed point

Use a smooth positive forcing (baseline prime density proxy)

$$g(n) = \frac{1}{\log(n+n_0)} \quad (n \geq 1), \quad g(0) = \frac{1}{\log(n_0+1)}.$$

Assuming $\max |\widehat{a}| < 1$, define the fixed point

$$f_\infty = (I-A)^{-1}g.$$

In Fourier space (DFT indices ξ):

$$\widehat{f}_\infty(\xi) = \frac{\widehat{g}(\xi)}{1 - \widehat{a}(\xi)}.$$

3. Layer 2: pattern-specific sieve mask

This layer implements the standard local admissibility viewpoint from sieve theory [3,4].

Let $H = \{h_1, \dots, h_k\}$ be a prime k -tuple pattern (e.g., $\{0, 2\}$ for twins). Fix a sieve cutoff y .

Define the truncated admissibility mask

$$M_{\{H,y\}}(n) = \prod_{p \leq y} \mathbf{1}_{\{\forall h \in H: n+h \not\equiv 0 \pmod p\}}.$$

Define the truncated Hardy–Littlewood singular series [1]

$$\mathfrak{S}_y(H) = \prod_{p \leq y} \frac{1 - \nu_H(p)/p}{(1-1/p)^k}$$

where $\nu_H(p)$ is the number of distinct residues occupied by H modulo p .

4. Observable (ratio form)

Define the weight signal

$$w(n) := \prod_{h \in H} f_{\infty}(n+h) \geq 0.$$

Define the primary observable

$$R_H(X; y) = \frac{\prod_{p \leq y} (1-1/p)^{-k}}{\sum_{n \leq X} M_{\{H,y\}}(n) w(n)} \frac{\sum_{n \leq X} M_{\{H,y\}}(n) w(n)}{\sum_{n \leq X} w(n)}.$$

This ratio is designed to remove slow log-scale variation in f_{∞} and isolate local admissibility structure.

5. Mathematical explanation of the negative result

5.1 Cancellation principle

The observable is a weighted density of the mask:

$$\frac{\sum_{n \leq X} M_{\{H,y\}}(n) w(n)}{\sum_{n \leq X} w(n)}.$$

If w is slowly varying relative to the periodic structure induced by primes $\leq y$, or approximately independent of $M_{H,y}$, then

$$\frac{\sum_{n \leq X} M_{\{H,y\}}(n) w(n)}{\sum_{n \leq X} w(n)} \approx \frac{1}{X} \sum_{n \leq X} M_{\{H,y\}}(n),$$

hence

$$R_H(X; y) \approx \mathfrak{S}_y(H).$$

This approximation becomes exact over full periods when w is constant on residue classes modulo the primorial $Q = \prod_{p \leq y} p$.

5.2 Why Layer 1 is generically “smooth” here

- Forcing g is positive and slowly varying.
- $(I - A)^{-1}$ is translation-invariant; its action is diagonal in Fourier space.
- With ρ -normalization, prime-support and random-support kernels produce similar-scale Fourier symbols $\hat{a}(\xi)$, so f_∞ stays in a low-frequency dominated, positive regime.
- In the ratio observable, most global reweighting cancels.

Consequence: k-tuple constant recovery is **sieve-dominated**, while Layer 1 is **arithmetic-agnostic** for these diagnostics.

6. Falsification protocols

We implement four tests intended to detect arithmetic content in Layer 1.

Test 1: Global constant recovery

Compare $R_H(X; y)$ to $\mathfrak{S}_y(H)$ for several patterns H .

Test 2: Residue-conditioned ratios (definitive)

For a modulus m , define residue-conditioned ratios

$$R_{H,a}(X; y; m) = \frac{\prod_{p \leq y} (1 - 1/p)^{-k}}{\sum_{n \leq X \atop n \equiv a \pmod{m}} M_{H,y}(n) w(n)} \Big/ \frac{\prod_{p \leq y} (1 - 1/p)^{-k}}{\sum_{n \leq X \atop n \equiv a \pmod{m}} w(n)}.$$

If Layer 1 encodes arithmetic structure, prime-support and random-support operators should produce measurably different vectors $(R_{H,a})_{a=0}^{m-1}$.

Test 3: Dirichlet character correlations

Probe periodic structure by computing correlations against a non-principal Dirichlet character χ [4] (e.g., mod 3):

$$C_\chi(X) = \sum_{n \leq X} f_\infty(n) \chi(n),$$

and compare prime-support vs random-support models.

Test 4: Frequency response signatures

Define

$$W(\lambda) = \sum_{p \leq P} p^\alpha e^{-2\pi i \lambda p}, \quad T(\lambda) = \frac{1}{|1 - \lambda W(\lambda)|}.$$

Compare $T(\xi)$ for prime support vs random support, including rational $\xi = a/q$ with small denominators and dense frequency grids. (See also standard spectral heuristics around zeta zeros, e.g. Montgomery's pair correlation [6].)

7. Numerical results (definitive outcomes)

All outcomes below are computed under a consistent parameterization used throughout the investigation.

7.1 Parameterization used in the definitive run

- $N = 2^{22} = 4,194,304$
- $X = 2,000,000$
- $P = 1000, y = 200$
- $\alpha = -1.5, \rho = 0.95, n_0 = 10$

7.2 Summary of falsification tests

- (1) **Global $**R_H**$:** matches $\mathfrak{S}_y(H)$ within $\sim 1\%$ (sieve dominance).
- (2) **Residue-conditioned ratios:** prime-support vs random-support vectors have correlation **1.0000** and $\text{MAE} < 3 \times 10^{-5}$ for mod 30 and mod 210.
- (3) **Dirichlet sums:** prime-support and random-support correlations are effectively identical (difference at $\sim 1\text{e-}10$ scale after normalization).
- (4) **Frequency response:** maximum observed z-scores lie within the noise envelope for many frequency bins; no statistically significant prime-specific enhancement after ρ -normalization.

8. Figures and data artifacts (reproducibility bundle)

Convergence and baseline comparison

- Full model convergence: `mscm2_corrected_full_model_convergence.png`
- f_∞ vs baseline density: `mscm2_corrected_f_vs_baseline.png`
- Summary table (CSV): `mscm2_corrected_summary.csv`
- Baseline vs full overlay: `mscm2_baseline_vs_full_overlay.png`

Residue-conditioned diagnostics

- Mod 30 scatter: `mscm2_residue_mod30_scatter.png`
- Mod 30 diff bars: `mscm2_residue_mod30_diffbar.png`
- Mod 210 scatter: `mscm2_residue_mod210_scatter.png`
- Mod 210 diff bars: `mscm2_residue_mod210_diffbar.png`
- Residue-conditioned summary (CSV): `mscm2_residue_conditioned_summary.csv`

Dirichlet diagnostic

- Dirichlet mod 3 summary (CSV): `mscm2_dirichlet_mod3_summary.csv`

Frequency-response diagnostics

- Rationals table (CSV): `mscm2_frequency_response_rationals.csv`
 - Ratio plot: `mscm2_frequency_response_ratio.png`
 - Z-scores at rationals: `mscm2_frequency_response_zscores.png`
 - Z-score histogram (grid): `mscm2_frequency_response_zscore_hist.png`
 - Z-score curve (grid): `mscm2_frequency_response_zscore_curve.png`
-

9. Interpretation and implications

9.1 Core conclusion

Within the tested model class,

Hardy-Littlewood constants emerge from local admissibility (sieve) structure [1,3,4].
Global prime-shift dynamics contributes no detectable arithmetic content to the studied ratio and spectral diagnostics.

9.2 What this rules out

- “Primes as eigenvalues” is **not supported** by translation-invariant convolution alone (cf. spectral heuristics around zeta zeros, e.g. Montgomery’s pair correlation [6]).
- Prime-indexed kernels do **not** automatically produce residue-class signatures.
- Spectral “resonances” at rationals do **not** appear as a robust prime-specific effect after stability normalization.

9.3 What this clarifies

This work identifies a broad dead-end: **linear translation-invariant operator models with smooth positive forcing** are unlikely to encode prime-specific structure in ratio observables.

This complements probabilistic models of primes such as Cramér’s model [2] and refinements discussed by Granville [5] by isolating the role of local admissibility.

10. Future directions (minimal changes that could matter)

To obtain arithmetic content in an operator model, at least one of the following must change.

Recent progress on bounded gaps and admissible k-tuple methods emphasizes the centrality of sieve structure [7,8].

1. **Break translation invariance** (introduce a residue-dependent potential $V(n)$):

$$(Af)(n) = \sum_{p \leq P} w(p) V(n) f(n-p).$$

1. **Use oscillatory forcing** (probe response without positivity/low-frequency cancellation):

$$g(n) = \chi(n) \quad \text{or} \quad g(n) = e^{2\pi i n/q}.$$

1. **Introduce nonlinearity:**

$$f_{t+1} = \sigma(Af_t + g),$$

where σ can amplify residue-dependent differences.

11. Paper-ready framing

Suggested title

Why Translation-Invariant Prime-Shift Operators Don’t Encode Arithmetic: A Negative Result for Dynamical Models of Prime k-Tuples

Suggested abstract

We study whether translation-invariant linear operators built from prime-indexed shifts can encode arithmetic information relevant to Hardy–Littlewood prime k-tuple correlations [1]. Using a two-layer architecture—(i) a global prime-shift convolution dynamics and (ii) a local pattern-specific sieve mask—we show that the Hardy–Littlewood constants are recovered entirely by the sieve layer [3,4]. Multiple falsification tests (global ratios, residue-conditioned ratios, Dirichlet character probes, and transfer-function diagnostics) demonstrate that the global operator is statistically indistinguishable from random-support analogues matched in density and stability (cf. probabilistic prime models [2,5]). We provide a mathematical explanation via ratio cancellation under smooth positive forcing and conclude that arithmetic content in operator models likely requires non-translation-invariance, oscillatory forcing, or nonlinear dynamics.

12. Checklist for final submission

-

13. References

- [1] G. H. Hardy and J. E. Littlewood, *Some Problems of 'Partitio Numerorum.'* III. *On the Expression of a Number as a Sum of Primes*, **Acta Mathematica** 44 (1923), 1–70. doi:10.1007/BF02403921.
- [2] H. Cramér, *On the order of magnitude of the difference between consecutive prime numbers*, **Acta Arithmetica** 2 (1936), 23–46. doi:10.4064/aa-2-1-23-46.
- [3] H. Halberstam and H.-E. Richert, *Sieve Methods*, Academic Press, 1974.
- [4] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications, Vol. 53, American Mathematical Society, 2004.
- [5] A. Granville, *Harald Cramér and the distribution of prime numbers*, **Scandinavian Actuarial Journal** 1995(1) (1995), 12–28. doi:10.1080/03461238.1995.10413946.
- [6] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in **Analytic Number Theory**, Proc. Sympos. Pure Math., Vol. 24, American Mathematical Society, 1973, pp. 181–193.
- [7] J. Maynard, *Small gaps between primes*, **Annals of Mathematics** 181(1) (2015), 383–413. doi:10.4007/annals.2015.181.1.7.
- [8] Y. Zhang, *Bounded gaps between primes*, **Annals of Mathematics** 179(3) (2014), 1121–1174. doi:10.4007/annals.2014.179.3.7.